

Using Normal-Model z-Scores and Percentiles

Answer Key

Precalculus / Statistics

Quick Answers

- | | |
|--|--|
| 1. 1.2 | 7. 1.5 |
| 2. -0.45 | 8. 16 |
| 3. 84 | 9. 46 hours |
| 4. 650 | 10. $\frac{1}{40}$ |
| 5. (a) z on the history test = 1.6, (b) z on the biology test = 1.5, — | 11. (a) the raw score = 63, — |
| 6. (a) the lower bound = 154 cm, (b) the upper bound = 186 cm | 12. (a) z for the first battery = 2, (b) z for the second battery = 1.4, — |
-

What is verified

15 of 18 answers machine-checked against SymPy, across 12 problems.

— marks an answer only you can judge — problems 5, 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus / Statistics

HSS-ID.A – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*

Difficulty 1–5 of 5 across 18 tagged checks.

Common wrong answers

- If they got 0.24: divided the deviation by the mean instead of by the standard deviation*
 - If they got 0.45: subtracted in the wrong order and lost the sign*
 - If they got 34: reported the percentage between the mean and one standard deviation below it instead of the percentage below that point*
-

1. Exam with $\mu = 500$, $\sigma = 100$. Find the z-score of 620.

Solution: $z = \frac{620 - 500}{100} = \frac{120}{100}.$

$z = 1.2$

2. Same exam. Find the z-score of 455.

Solution: $z = \frac{455 - 500}{100} = \frac{-45}{100}$. The negative sign says the score is below the mean.

$$z = -0.45$$

3. Same exam. What percentage of scores fall below 600?

Solution: 600 is exactly 1σ above the mean. The empirical rule puts 68% within 1σ , so 34% sits between the mean and 600, and 50% sits below the mean: $50 + 34$.

$$84$$

4. Same exam. A student's z -score is 1.5. Find their score.

Solution: $x = \mu + z\sigma = 500 + 1.5(100) = 500 + 150$.

$$x = 650$$

5. History: 88 with $\mu = 80$, $\sigma = 5$. Biology: 76 with $\mu = 70$, $\sigma = 4$. (a) and (b) the z -scores. (c) which was stronger?

Solution (a): $z = \frac{88 - 80}{5} = \frac{8}{5}$.

$$z = 1.6$$

Solution (b): $z = \frac{76 - 70}{4} = \frac{6}{4}$.

$$z = 1.5$$

Solution (c) — instructor-judged. The history result is 1.6 standard deviations above its class mean; the biology result is 1.5 above its own. Relative to the class, the history performance is very slightly stronger. The raw scores cannot settle it, because 88 and 76 are measured against different means (80 versus 70) and different spreads (5 versus 4) — standardizing is what makes them comparable. Full credit requires the comparison of z -scores and the reason the raw scores are not comparable.

6. Heights: $\mu = 170$ cm, $\sigma = 8$ cm. About 95% lie between two values. (a) lower. (b) upper.

Solution (a): 95% corresponds to 2σ each side: $170 - 2(8) = 170 - 16$.

$$154 \text{ cm}$$

Solution (b): $170 + 2(8) = 170 + 16$.

$$186 \text{ cm}$$

7. Same heights. Find the z -score of 182 cm.

Solution: $z = \frac{182 - 170}{8} = \frac{12}{8}$.

$$z = 1.5$$

8. Same heights. What percentage is shorter than 162 cm?

Solution: 162 is 1σ below the mean. Of the 68% within 1σ , half (34%) lies between 162 and the mean, and 50% lies below the mean, so the percentage below 162 is $50 - 34$.

$$16$$

9. Batteries: $\mu = 40$ hours, $\sigma = 3$ hours. Find the 97.5th percentile.

Solution: 97.5% below means 2.5% above, which is one tail of the 95% band, so $z = 2$:
 $x = 40 + 2(3) = 40 + 6$. 46 hours

10. Same batteries: 25 in every 1000 last longer than the 97.5th percentile. Give the probability as a fraction.

Solution: $P = \frac{25}{1000}$, and dividing both parts by 25 gives $\frac{1}{40}$ — the 2.5% upper tail. $P = \frac{1}{40}$

11. Test with $\mu = 72$, $\sigma = 6$; a student's z -score is -1.5 . (a) raw score. (b) what the sign means and the percentile.

Solution (a): $x = 72 + (-1.5)(6) = 72 - 9$. $x = 63$

Solution (b) — instructor-judged. A negative z means the score is below the mean — here 1.5 standard deviations below. For the percentile: about 32% of the distribution lies more than 1σ from the mean and about 5% more than 2σ , so the lower tail beyond 1.5σ holds roughly 6–7% of the values. The student is near the 7th percentile. Full credit for "below the mean" plus any percentile estimate between about 5 and 10 with a stated reason.

12. Synthesis. Line 1: $\mu = 40$, $\sigma = 3$, one battery at 46 hours. Line 2: $\mu = 45$, $\sigma = 5$, one battery at 52 hours. (a), (b) the z -scores. (c) which is more unusual?

Solution (a): $z = \frac{46 - 40}{3} = \frac{6}{3}$. $z = 2$

Solution (b): $z = \frac{52 - 45}{5} = \frac{7}{5}$. $z = 1.4$

Solution (c) — instructor-judged. The Line 1 battery is 2 standard deviations above its own mean; the Line 2 battery only 1.4 above its own. The Line 1 result is the more unusual, even though the Line 2 battery lasted 6 hours longer in absolute terms. Comparing 46 with 52 directly is meaningless here because the two lines have different means and different spreads — Line 2 is both better on average and more variable, so a long life there is less surprising. Full credit requires the z -comparison and an explanation of why the raw lifetimes mislead.